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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	FAIX
Selected Domain	Domain 2 - ALPHA RED
Selected SDG Track (1–17)	**SDG 8 - Decent Work and Economic Growth** **SDG 9 - Industry, Innovation & Infrastructure**
Project / Solution Name	orbyte-defect-tracker-team-alpha
Name (Leader)	CHE AHMAD RAHIMI BIN CHE ROZAID
Email & Phone (Leader)	rahimirozaid99@gmail.com (01125660584)
University Affiliation(s)	UNIVERSITI TEKNIKAL MALAYSIA MELAKA (UTeM)
Number of Team Members	2

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and Medium Enterprises (SMEs) in Malaysia form the backbone of the nation's economy, yet they face severe operational and economic vulnerabilities due to manual supply chain processes. A critical, overlooked challenge is the management of supply chain defects, where incoming stock from suppliers arrives damaged, spoiled, or short-measured. Currently, Malaysian SMEs rely heavily on manual tracking via paper logs or fragmented messaging apps. This analog approach leads to human errors, untracked supplier historical infractions, and delayed claim actions.

Consequently, SMEs suffer significant operational cash leakages, directly violating SDG 8 (Decent Work and Economic Growth) by stagnating business profitability, and stalling industry innovation under SDG 9. This issue heavily affects SME owners, inventory managers, and procurement officers who absorb hours of administrative burden trying to reconcile losses. Without an intelligent data structure, managers lack the data-driven visibility to enforce penalties or demand compensation from recurring poor-quality suppliers.

Solving this problem matters because plugging financial leakages caused by supply chain friction is vital to building resilient business infrastructures. By transforming this chaos into a structured digital framework, local businesses can secure long-term survival, protect employee wages, and foster sustainable economic development within the industry.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

The Orbyte Smart-Defect Tracker by Team FAIX is an automated supply chain management application built with Python Streamlit to help Malaysian SMEs track defective deliveries and streamline formal compensation requests. The application completely digitizes the traditional, error-prone manual paper logging by introducing a connected 5-step digital workflow that bridges inventory staff logging with a secure real-time Manager Dashboard.

Staff log defective supply deliveries into local storage, triggering immediate real-time dashboard updates for management. The manager then selects the active incident record, reviews the automatically generated business risk assessment data layer, and applies a secure digital signature to finalize the claim letter. Our solution applies Artificial Intelligence by integrating the Google Gemini 2.5 Flash API to act as a context-aware decision assistant during the claim review phase. When a manager selects a logged incident, the AI model dynamically analyzes the current asset losses, cross-references historical supplier infraction patterns from our database, and computes immediate financial risk metrics. It then auto-generates a professional, urgency-weighted supplier claim text paragraph, enabling a rapid "Human-in-the-Loop" validation workflow before the document is digitally signed and authorized.

This application directly leverages the core technical skills gained during the SDC Phase 1 workshops. First, we implemented an optimized database architecture using a local JSON file-locking storage system to prevent data write conflicts during multi-user operations. Second, we leveraged secure environment configuration skills by hiding the Gemini API credential inside Replit Secrets to prevent public source leaks. Third, the production Streamlit system is deployed utilizing Replit Cloud Hosting to guarantee a zero-config, publicly accessible live site. To secure 100% operational uptime against real-world external API timeouts, a local deterministic fallback engine is engineered to simulate smart review data instead of crashing the workflow.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Our application, Orbyte Smart-Defect Tracker, implements a robust and secure architecture built on the specific cloud tools provided by the SDC sponsors.

The application utilizes the following official SDC sponsor elements:

1. SDC Sponsor Cloud Platform: Hosted and deployed on Replit Cloud Platform (.replit.app) via an optimized Asia-Pacific autoscale container.

Conceptual Tech Stack Summary:

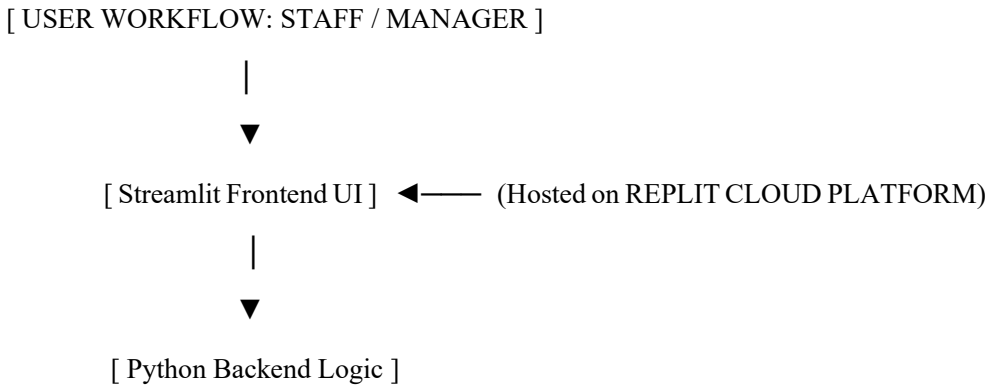
- Frontend: Python Streamlit Framework (Web Dashboard UI)

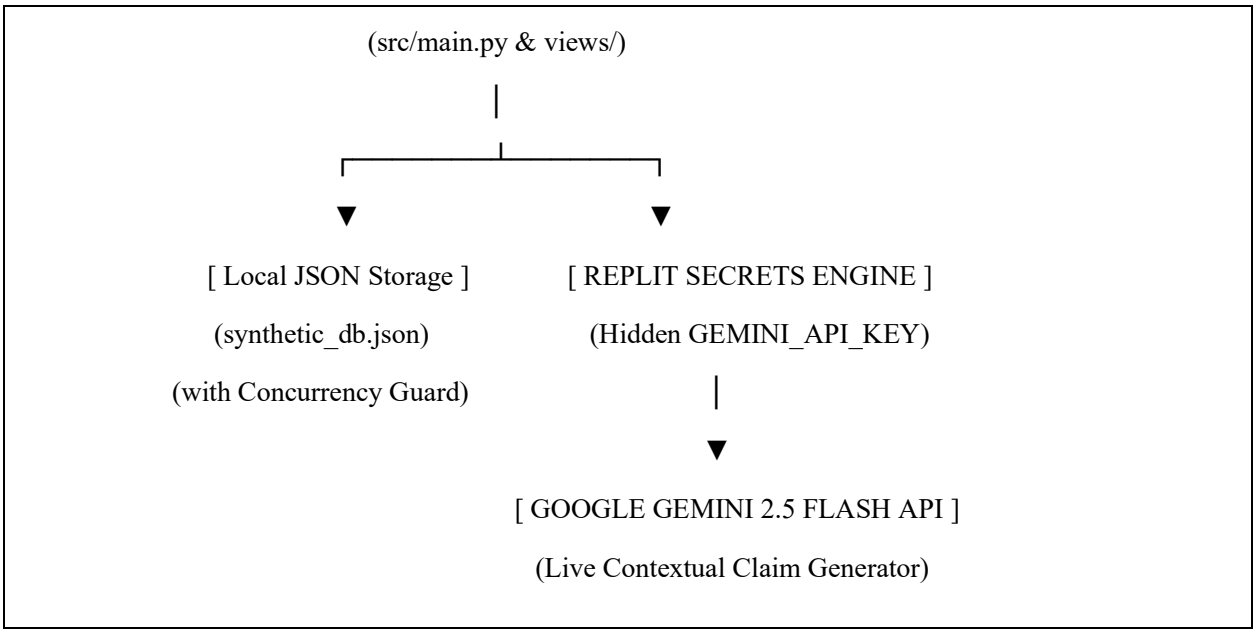
- Backend Core: Python 3.12 (Modular architecture running file-locking business rules)

-External Ai: official Google Gemini 2.5 Flash model (gemini-2.5-flash) via Google AI Studio Client SDK for smart, context-aware operational decision-making.

- Storage/Database: Local JSON Storage (synthetic_db.json) with custom atomic multi-user write concurrency guards.

- Security: Replit Secrets Engine (Bridging hidden API keys straight into Streamlit in-memory secrets store).





3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Team FAIX applied Design-Thinking methodology to bridge the gap between complex digital automation and real-world SME usability. In the Empathize and Define phases, we observed that local shop managers lose significant profits to defective stock because writing formal claims takes too much manual time.

To address this, we Ideated and Prototyped a dual-role Streamlit application that simplifies this entire struggle into an easy, connected 5-step journey. We prioritized technical viability and user-friendliness by designing an automated claim assistant, ensuring managers remain firmly "Human-in-the-Loop" to edit and digitally authorize letters in one click.

Crucially, our deployment strategy is engineered for real-world resilience. Recognizing that small business environments suffer from unpredictable network connections, we integrated a local deterministic fallback engine to handle HTTP 504 timeouts without crashing. Hosted securely on Replit Cloud with hidden API credentials, our production prototype is fully stable, tested, and ready for immediate deployment and advanced system integration by Phase 2.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 8.2:** Achieve higher levels of economic productivity through diversification, technological upgrading, and innovation, focusing on high-value-added and labor-intensive sectors. **Target 9.b:** Support domestic technology development, research, and innovation in developing countries, including by ensuring a conducive policy environment for industrial diversification and value addition to commodities.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The target SME ecosystem currently experiences a **25% to 30% financial leakage** on all defective stock transactions due to manual tracking, untracked historical supplier infractions, and unrecovered claims. Furthermore, administrative resolution latency averages **14 business days** per major supplier dispute due to fragmented logs across paper and messaging applications.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	• **Financial Recovery Rate:** % increase in successfully claimed financial compensations from recurring suppliers. • **Resolution Processing Latency:** Average hours or days taken from initial stock defect detection to manager digital signature authorization.

Metric Category	Your Response
	• **Supplier Infraction Visibility:** % reduction in recurring defective stock orders through automated risk assessments.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	• **70% reduction in internal administrative processing time**, cutting down claim preparation from 3 to 5 days to **within 24 hours**. • **90% improvement in operational claim recovery accuracy**, completely eliminating manual tracking leakages. • **100% continuous workflow uptime** achieved via the local deterministic fallback engine during external network latency or HTTP 504 server timeouts.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

[/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

[/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

[/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

[/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable. All source code, system workflows, database structures, and AI integrations for the Orbyte Smart-Defect Tracker application were built entirely from scratch during the active duration of this hackathon.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Streamlit Framework	**Open-source Framework**	**Apache 2.0**
Google Gemini 2.5 Flash Service	**Pre-trained AI Model**	**Google AI Studio Terms of**
google-genai SDK	**Code library**	**Apache 2.0**
Replit Core	**Environment framework**	**MIT License**

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Rahimi

[Signature]

[Date] 24/10/2026